

```
1 // LAB CODE: 09876
2 //=====
3 //Jason Johnson
4 //CS1A 1:30 - 3:50
5 //Med Mogasemi
6 // 11/12/25
7 //=====
8
9
10 #include <iostream>
11 #include <iomanip>
12
13 using namespace std;
14
15 void getScore(double& score1, double& score2, double& score3, double& score4, double& score5);
16 double calcAverage(double score1, double score2, double score3, double score4, double score5);
17 char determineGrade(double avg);
18 void displayGrade(char grade);
19
20 int main() {
21     double score1{};
22     double score2{};
23     double score3{};
24     double score4{};
25     double score5{};
26     double avg{};
27     char grade{};
28
29     getScore(score1, score2, score3, score4, score5);
30
31     avg = calcAverage(score1, score2, score3, score4, score5);
32
33     grade = determineGrade(avg);
34
35     displayGrade(grade);
36
37
38     return 0;
39 }
40
41
42
43
44 void getScore(double& score1, double& score2, double& score3,
45             double& score4, double& score5) {
46
47     cout << "Please enter 5 scores: \n\n";
```

```
48
49     cout << "Score 1: ";
50     cin >> score1;
51
52     cout << "Score 2: ";
53     cin >> score2;
54
55     cout << "Score 3: ";
56     cin >> score3;
57
58     cout << "Score 4: ";
59     cin >> score4;
60
61     cout << "Score 5: ";
62     cin >> score5;
63
64     cout << endl;
65
66     while (score1 < 0 || score1 > 100 || score2 < 0 || score2 > 100
67           || score3 < 0 || score3 > 100 || score4 < 0 || score4 > 100
68           || score5 < 0 || score5 > 100)
69     {
70         cout << "INVALID DATA. Please enter a score larger"
71              << " than 0 and not greater than 100\n\n";
72
73         cout << "Please enter 5 scores: \n\n";
74
75         cout << "Score 1: ";
76         cin >> score1;
77
78         cout << "Score 2: ";
79         cin >> score2;
80
81         cout << "Score 3: ";
82         cin >> score3;
83
84         cout << "Score 4: ";
85         cin >> score4;
86
87         cout << "Score 5: ";
88         cin >> score5;
89
90     }
91
92 }
93
94
95
96
```

```
97
98 double calcAverage(double score1, double score2, double score3, double  ➤
    score4, double score5) {
99     return (score1 + score2 + score3 + score4 + score5) / 5;
100 }
101
102
103
104
105 char determineGrade(double avg) {
106
107     if (avg > 90) {
108         return 'A';
109     }
110     else if (avg > 80) {
111         return 'B';
112     }
113     else if (avg > 70) {
114         return 'C';
115     }
116     else if (avg > 60) {
117         return 'D';
118     }
119     else
120         return 'F';
121
122 }
123
124
125
126 void displayGrade(char grade) {
127     cout << "Your Grade is: " << grade;
128     cout << endl;
129 }
130
131
132
133
134
135 //=====
136 // Sample Output:
137 /*
138
139 Please enter 5 scores:
140
141 Score 1: 90
142 Score 2: 80
143 Score 3: 77
144 Score 4: 85
```

```
145 Score 5: 72
146
147 Your Grade is: B
148
149 C:\Users\jjohnson257\source\repos\Project13\x64\Debug\
150 Project13.exe (process 22224) exited with code 0 (0x0).
151 Press any key to close this window . . .
152
153
154 or
155
156
157 Please enter 5 scores:
158
159 Score 1: 0
160 Score 2: 0
161 Score 3: 0
162 Score 4: 0
163 Score 5: 200
164
165 INVALID DATA. Please enter a score larger than 0 and not greater than 100
166
167 Please enter 5 scores:
168
169 Score 1: 20
170 Score 2: 30
171 Score 3: 40
172 Score 4: 50
173 Score 5: 60
174 Your Grade is: F
175
176 C:\Users\jjohnson257\source\repos\Project13\x64\Debug\Project13.exe
    (process 12188) exited with code 0 (0x0).
177 Press any key to close this window . . .
178
179
180
181
182 */
```